

BANK TRANSACTION ANALYTICS

SQL Project Report

Global Trust Bank

A structured SQL analytics project covering customer transactions, business KPIs, branch performance, customer analytics, and risk/fraud analysis.

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SQL • Excel • Python • Analytics

Project Report

1. Executive Summary

Bank Transaction Analytics is a SQL-based business analytics project designed to help Global Trust Bank understand transaction activity and customer behavior. The project applies more than 100 SQL queries to a banking transaction dataset and organizes the analysis into reporting, business KPIs, customer analytics, branch performance, risk analytics, and dashboard KPIs.

The project demonstrates how transactional data can be transformed into useful business information. It examines deposits and withdrawals, transaction values, customer segments, branch activity, payment channels, geographic performance, and potential risk indicators.

Important: This report summarizes the analytical framework and intended business insights from the supplied project README. Numerical KPI results are not stated because the underlying SQL output/data was not provided.

2. Project Objectives

The primary objective is to build a practical banking analytics solution using SQL and answer business questions that support reporting and decision-making.

Objective	Description
Transaction Analysis	Understand deposits, withdrawals, transaction values and trends.
Customer Analytics	Identify customer segments, high-value customers and behavioral patterns.
Branch Analytics	Compare branch activity and business performance.
Risk Analytics	Identify potential risk and fraud indicators using transaction and credit information.
KPI Reporting	Create reusable measures for dashboards and management reporting.
Trend Analysis	Analyze monthly, quarterly and yearly transaction patterns.

3. Database Overview

The project uses a MySQL database named **BankDB** with a central table **bank_transactions**. The table combines transaction-level information with customer, branch, geographic, payment, credit and loan attributes.

3.1 Main Table: bank_transactions

Category	Columns
Transaction	transaction_id, transaction_date, transaction_type, transaction_amount, balance_after_transaction
Customer	customer_id, customer_name, gender, age, occupation, customer_segment
Location	city, state, branch_name
Account	account_type, loan_status
Payment	payment_channel
Risk / Credit	credit_score

4. SQL Analysis Modules

Module	Queries	Focus
Module 1	1–20	Basic reporting and transaction-level exploration
Module 2	21–40	Business KPIs and financial measures
Module 3	41–60	Customer analytics and segmentation
Module 4	61–75	Branch performance analysis
Module 5	76–90	Risk and fraud analytics
Module 6	91–100	Dashboard KPI preparation

5. Business Questions

The SQL modules are designed to answer practical questions such as:

- Which branch generates the highest deposits?
- Which customers are high-value based on business above ■5,00,000?
- Which customers show high-risk characteristics based on low credit score and frequent withdrawals?
- Which city or state contributes the most business?
- Which payment channel is most frequently used?
- What are the monthly and quarterly transaction trends?
- Which customer segments dominate transaction activity?
- Which branches show growth or decline in performance?

6. Key Business KPIs

KPI	Business Purpose
Total Business	Measures overall transaction business handled by the dataset.
Total Deposits	Tracks deposit transaction value.
Total Withdrawals	Tracks withdrawal transaction value.
Average Transaction Value	Shows the typical monetary size of a transaction.
Highest Transaction	Identifies the maximum transaction amount.
Lowest Transaction	Identifies the minimum transaction amount.
City-wise Business	Compares geographic contribution.
Branch-wise Performance	Compares business activity across branches.
Customer Segment Analysis	Examines Premium, Business, Salary and other segments.
Fraud & Risk Flags	Highlights transactions/customers requiring further review.
Monthly / Quarterly / Yearly Trends	Supports time-based performance monitoring.

7. SQL Concepts Demonstrated

The project progresses from foundational SQL operations to business-oriented analytics. The README identifies the following techniques:

- Database creation and table design
- SELECT, WHERE, ORDER BY and DISTINCT
- LIMIT, LIKE, BETWEEN and IN
- Aggregate functions: SUM, AVG, MAX, MIN and COUNT
- GROUP BY and HAVING
- Date functions including MONTH(), QUARTER() and YEAR()
- Business KPI reporting
- Customer and branch analytics
- Fraud and risk detection queries

8. Analytical Insights Framework

Because the supplied README does not include the actual query output, the following are the types of insights the completed analysis is intended to produce rather than claimed numerical findings.

Area	Expected analytical insight
Deposits	Identify branches and customer groups contributing the greatest deposit activity.
Withdrawals	Understand withdrawal concentration and unusual withdrawal behavior.
Customers	Separate high-value customers from lower-activity groups using transaction measures.
Geography	Compare business contribution across cities and states.
Payment Channels	Measure adoption of UPI, ATM, Cheque, Net Banking and Cash.
Branches	Identify branches with stronger or weaker transaction activity over time.
Risk	Flag combinations of credit and transaction behavior that warrant investigation.
Time Trends	Identify seasonal, monthly and quarterly changes in transaction activity.

9. Repository Structure

File	Purpose
schema.sql	Database and table creation with sample data
module1_basic_reporting.sql	Queries 1–20: basic reporting
module2_business_kpis.sql	Queries 21–40: business KPIs
module3_customer_analytics.sql	Queries 41–60: customer analytics
module4_branch_performance.sql	Queries 61–75: branch performance
module5_risk_analytics.sql	Queries 76–90: fraud and risk analytics

File	Purpose
module6_dashboard_kpis.sql	Queries 91–100: dashboard KPIs
report.pdf	Project insights and KPI report

10. Project Workflow

Step 1: Import schema.sql into MySQL to create the BankDB database and bank_transactions table.

Step 2: Execute the six SQL modules in sequence to perform reporting, KPI, customer, branch, risk and dashboard analysis.

Step 3: Review query outputs and convert the most useful KPIs into dashboard visuals using Excel, Power BI or Tableau.

Step 4: Use the results to support business reporting, customer analysis, branch monitoring and risk review.

11. Skills Demonstrated

- SQL query writing and optimization
- Business data analysis
- KPI development and reporting
- Customer segmentation and analytics
- Branch performance analysis
- Fraud detection and risk analytics
- Database design and management
- Real-world application of SQL in banking analytics

12. Conclusion

The Bank Transaction Analytics project demonstrates how SQL can be used to convert transaction-level banking data into a structured analytical solution. With more than 100 queries across six modules, the project covers the complete path from basic data reporting to business KPIs, customer and branch analysis, risk indicators and dashboard-ready metrics.

The solution is also suitable as a foundation for a visual analytics dashboard in Excel, Power BI or Tableau. Adding the actual query outputs would allow the report to present numerical KPI values, charts and data-backed findings.

13. Future Scope

- Build an interactive Power BI or Tableau dashboard.
- Add automated KPI refresh and scheduled reporting.
- Introduce advanced fraud detection rules and anomaly scoring.
- Use SQL window functions and CTEs for deeper analysis.
- Integrate customer-level historical trends.
- Add predictive analytics using Python or machine learning.

- Create role-based dashboards for management, branch teams and risk analysts.

14. Author

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Passionate about SQL, Excel, Python and analytics projects, with a focus on building neat, structured and professional portfolio projects.